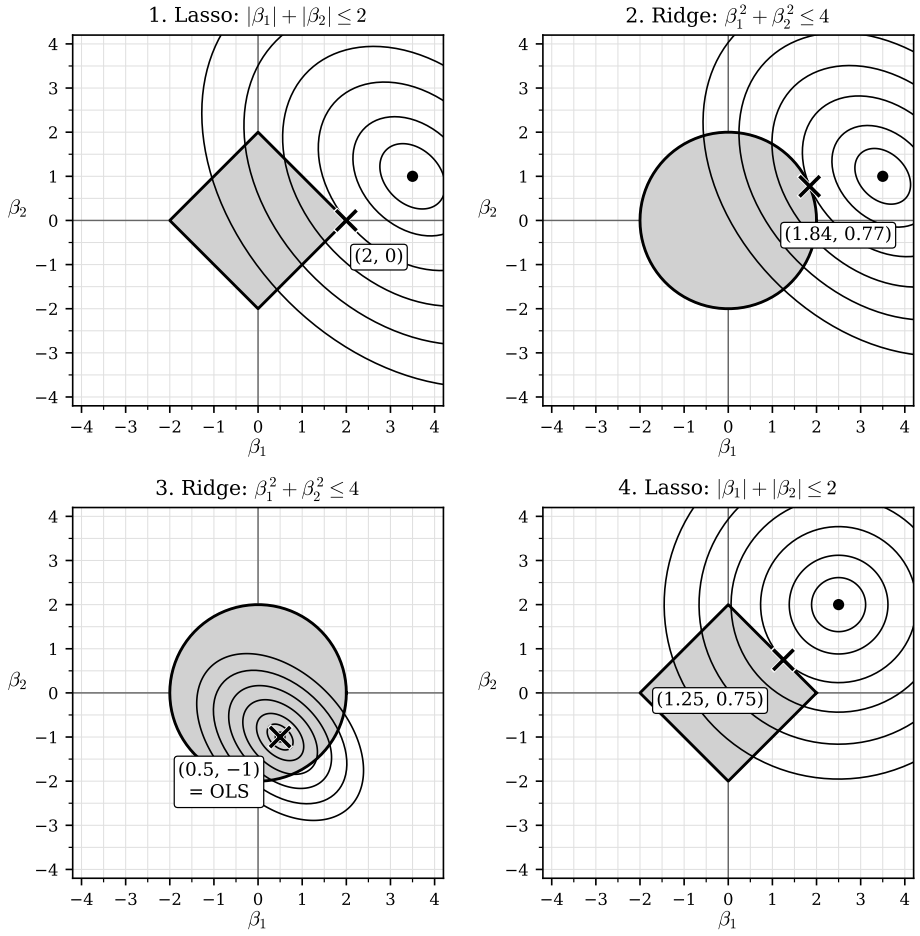
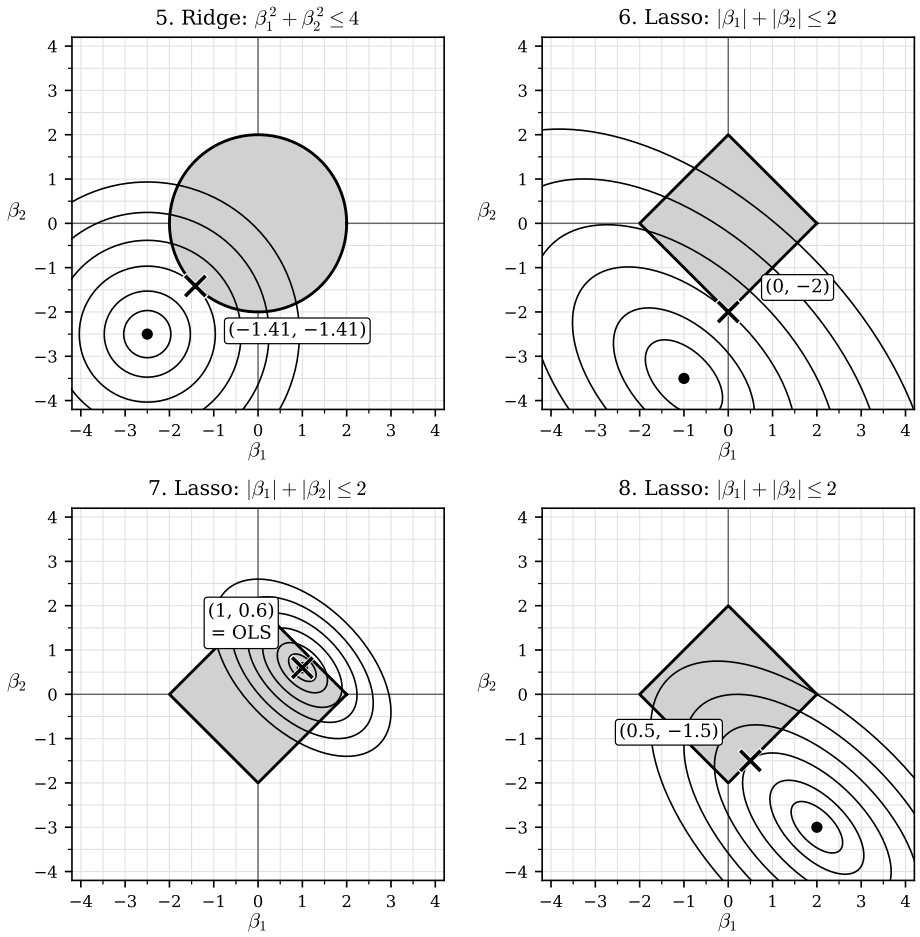
Intro to Machine Learning
Quiz 2 Practice: Constraint Pictures (Solutions)

Each panel shows contours of the RSS for a model with two predictors (ellipses around the OLS estimate, the solid dot) and a shaded hard-constraint region: the lasso, $|\beta_1| + |\beta_2| \le 2$, or ridge, $\beta_1^2 + \beta_2^2 \le 4$. The constrained estimate $(\hat{\beta}_1, \hat{\beta}_2)$ is the point of the shaded region with the smallest RSS.
For each panel: (a) mark the constrained estimate with an X; (b) read off its approximate coordinates (the grid lines are every 0.5); (c) say whether either coefficient is exactly 0.
Hint: RSS is constant along each ellipse and grows as the ellipses get bigger. Not every contour is drawn, so imagine the ones in between.



1. $(2, 0)$, so $\hat{\beta}_2 = 0$. The contours first reach the diamond at its right corner. The ellipses are nearly circles and OLS is far to the right with β_2 small, so the corner on the β_1 axis is the closest point.
2. About $(1.84, 0.77)$; neither is 0. Same RSS as panel 1, but the circle has no corners, so the first contact is on a smooth part of the boundary. Ridge shrinks both coefficients toward 0 and sets neither to 0.
3. $(0.5, -1.0)$, the OLS estimate; neither is 0. OLS is inside the disc ($0.5^2 + 1^2 = 1.25 \le 4$), so the constraint does not bind and nothing is shrunk.
4. $(1.25, 0.75)$; neither is 0. The contours are circles, so the estimate is the point of the diamond closest to $(2.5, 2.0)$: straight down onto the edge $\beta_1 + \beta_2 = 2$, along the direction $(1, 1)$ perpendicular to it. OLS is close to the diagonal, so the lasso lands on an edge, not a corner.



5. About $(-1.41, -1.41)$, that is $(-\sqrt{2}, -\sqrt{2})$; neither is 0. The contours are circles, so ridge pulls OLS straight toward the origin until it reaches the circle of radius 2.
6. $(0, -2)$, so $\hat{\beta}_1 = 0$. OLS is mostly in the β_2 direction, and the contours first reach the diamond at its bottom corner. This is how the lasso does variable selection.
7. $(1.0, 0.6)$, the OLS estimate; neither is 0. $|1.0| + |0.6| = 1.6 \le 2$, so OLS is already inside the diamond. A lasso constraint that does not bind sets nothing to 0.
8. $(0.5, -1.5)$; neither is 0. The two predictors are strongly correlated, so the ellipses are long and tilted, and the first contact is on the lower-right edge, not at the corner $(0, -2)$ that is closest in plain distance. Look at the shape of the contours, not only at where OLS is.

How to find the estimate.

- If OLS is inside the region, the constraint does not bind and the estimate is OLS.
- Otherwise, grow the ellipses out from OLS; the estimate is the first point of the region they touch, where a contour is tangent to the boundary (or passes through a corner).
- Ridge: the circle has no corners, so a coefficient is exactly 0 only by coincidence.
- Lasso: the estimate is often at a corner, where one coefficient is exactly 0, but not always (panels 4 and 8).